

FINITE AUTOMATA AND REGULAR LANGUAGE

(1 MARKS)

1. GATE CSE 2015 Set 3

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let L be the language represented by the regular expression $\sum^*0011\sum^*$ where $\sum = \{0, 1\}$. What is the minimum number of states in a DFA that recognizes $\overline{L}$ (complement of L)?

- (A) 4
- (B) 5
- (C) 6
- (D) 8

2. GATE CSE 2026 Set 1

MCQ (More than One Correct Answer) Marks: +1 / -0

Let M be a non-deterministic finite automaton (NFA) with 6 states over a finite alphabet. Which of the following options CANNOT be the number of states in the minimal deterministic finite automaton (DFA) that is equivalent to M ?

- (A) 32
- (B) 65
- (C) 1
- (D) 128

3. GATE CSE 2025 Set 1

MCQ (Single Correct Answer) Marks: +1 / -0

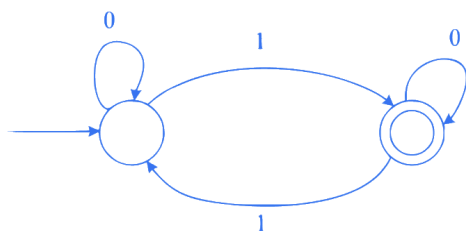
A regular language L is accepted by a non-deterministic finite automaton (NFA) with n states. Which of the following statement(s) is/are FALSE?

- (A) L may have an accepting NFA with $< n$ states.
- (B) L may have an accepting DFA with $< n$ states.
- (C) There exists a DFA with $\leq 2^n$ states that accepts L .
- (D) Every DFA that accepts L has $> 2^n$ states.

4. GATE CSE 2024 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.33

Which one of the following regular expressions is equivalent to the language accepted by the DFA given below?



- (A) $0^*1(0 + 10^*1)^*$
- (B) $0^*(10^*11)^*0^*$
- (C) $0^*1(010^*1)^*0^*$
- (D) $0(1 + 0^*10^*1)^*0^*$

5. GATE CSE 2024 Set 1

MCQ (More than One Correct Answer) Marks: +1 / -0

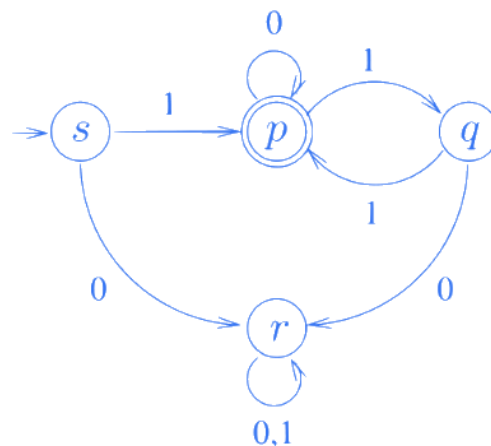
Let L_1, L_2 be two regular languages and L_3 a language which is not regular. Which of the following statements is/are always TRUE?

- (A) $L_1 = L_2$ if and only if $L_1 \cap \overline{L_2} = \emptyset$
- (B) $L_1 \cup L_3$ is not regular
- (C) $\overline{L_3}$ is not regular
- (D) $L_1 \cup \overline{L_2}$ is regular

6. GATE CSE 2023

MCQ (Single Correct Answer) Marks: +1 / -0.33

Consider the Deterministic Finite-state Automation (DFA) A shown below. The DFA runs on the alphabet $\{0, 1\}$, and has the set of states $\{s, p, q, r\}$, with s being the start state and p being the only final state.



Which one of the following regular expressions correctly describes the language accepted by A ?

- (A) $1(0^*11)^*$
- (B) $0(0 + 1)^*$
- (C) $1(0 + 11)^*$
- (D) $1(110^*)^*$

7. GATE CSE 2023

MCQ (Single Correct Answer) Marks: +1 / -0.33

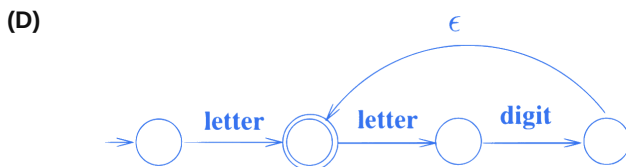
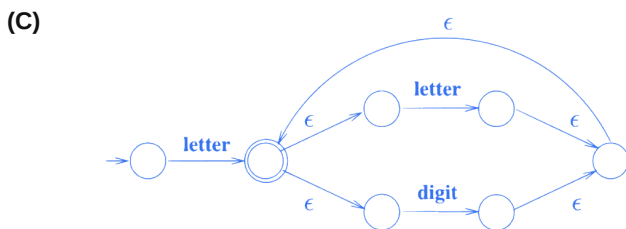
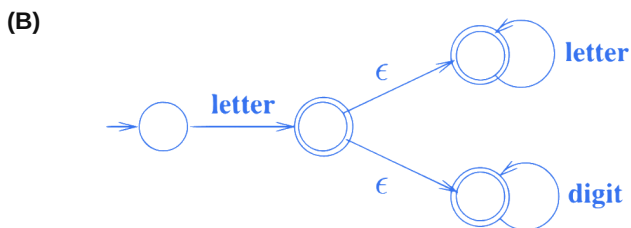
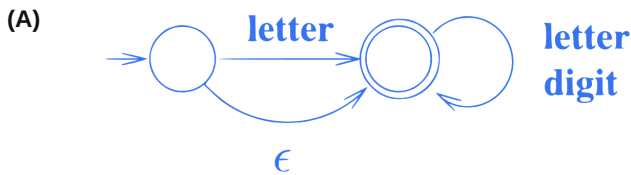
Consider the following definition of a lexical token id for an identifier in a programming language, using extended regular expressions:

letter $\rightarrow [A - Z a - z]$

letter $\rightarrow [0 - 9]$

id $\rightarrow \text{letter}(\text{letter} \mid \text{digit})^*$

Which one of the following Non-deterministic Finite-state Automata with ϵ -transmissions accepts the set of valid identifiers? (A double-circle denotes a final state)



8. GATE CSE 2023

MCQ (More than One Correct Answer) Marks: +1 / -0

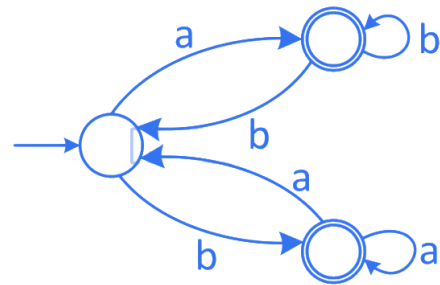
Which of the following statements is/are CORRECT?

- (A) The intersection of two regular languages is regular.
- (B) The intersection of two context-free languages is context-free.
- (C) The intersection of two recursive languages is recursive.
- (D) The intersection of two recursively enumerable languages is recursively enumerable.

9. GATE CSE 2022

MCQ (Single Correct Answer) Marks: +1 / -0.33

Which one of the following regular expressions correctly represents the language of the finite automaton given below?



- (A) $ab^*bab^* + ba^*aba^*$
- (B) $(ab^*b)^* ab^* + (ba^*a)^* ba^*$
- (C) $(ab^*b + ba^*a)^* (a^* + b^*)$
- (D) $(ba^*a + ab^*b)^* (ab^* + ba^*)$

10. GATE CSE 2021 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.33

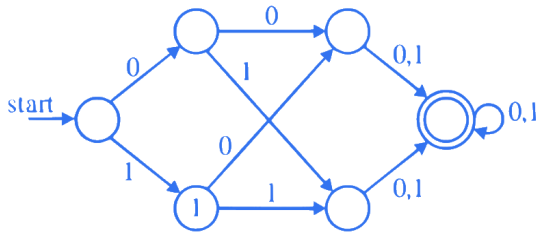
Let $L \subseteq \{0,1\}^*$ be an arbitrary regular language accepted by a minimal DFA with k states. Which one of the following languages must necessarily be accepted by a minimal DFA with k states?

- (A) $\{0,1\}^* - L$
- (B) $L.L$
- (C) $L - \{01\}$
- (D) $L \cup \{01\}$

11. GATE CSE 2021 Set 2

Numerical Marks: +1 / -0

Consider the following deterministic finite automaton (DFA).



The number of strings of length 8 accepted by the above automaton is _____

12. GATE CSE 2020

MCQ (Single Correct Answer) Marks: +1 / -0.33

Consider the following statements.

I. If $L_1 \cup L_2$ is regular, then both L_1 and L_2 must be regular.**II.** The class of regular languages is closed under infinite union.Which of the above statements is/are **TRUE**?

- (A) I only
 (B) II only
 (C) Both I and II
 (D) Neither I nor II

13. GATE CSE 2020

MCQ (Single Correct Answer) Marks: +1 / -0.33

Which one of the following regular expressions represents the set of all binary strings with an odd number of 1's?

- (A) $10^*(0^*10^*10^*)^*$
 (B) $((0+1)^*1(0+1)^*1)^*10^*$
 (C) $(0^*10^*10^*)^*10^*$
 (D) $(0^*10^*10^*)^*0^*1$

14. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +1 / -0.33

If L is a regular language over $\Sigma = \{a, b\}$, which one of the following languages is NOT regular?

- (A) Suffix $(L) = \{y \in \Sigma^* \text{ such that } xy \in L\}$
 (B) $\{ww^R \mid w \in L\}$
 (C) Prefix $(L) = \{x \in \Sigma^* \mid \exists y \in \Sigma^* \text{ such that } xy \in L\}$
 (D) $L \cdot L^R = \{xy \mid x \in L, y^R \in L\}$

15. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +1 / -0.33

For $\Sigma = \{a, b\}$, let us consider the regular language $L = \{x \mid x = a^{2+3k} \text{ or } x = b^{10+12k}, k \geq 0\}$. Which one of the following can be a pumping length (the constant guaranteed by the pumping lemma) for L ?

- (A) 9
 (B) 5
 (C) 24
 (D) 3

16. GATE CSE 2016 Set 1

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which one of the following regular expressions represents the language: the set of all binary strings having two consecutive 0s and two consecutive 1s?

- (A) $(0+1)^*0011(0+1)^* + (0+1)^*1100(0+1)^*$
 (B) $(0+1)^*(00(0+1)^*11 + 11(0+1)^*00)(0+1)^*$
 (C) $(0+1)^*00(0+1)^* + (0+1)^*11(0+1)^*$
 (D) $00(0+1)^*11 + 11(0+1)^*00$

17. GATE CSE 2016 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.3

Language L_1 is defined by the grammar: $S_1 \rightarrow aS_1b \mid \varepsilon$ Language L_2 is defined by the grammar: $S_2 \rightarrow abS_2 \mid \varepsilon$

Consider the following statements:

 $P : L_1$ is regular $Q : L_2$ is regularWhich one of the following is **TRUE**?

- (A) Both P and Q are true
 (B) P is true and Q is false
 (C) P is false and Q is true
 (D) Both P and Q are false

18. GATE CSE 2016 Set 2

Numerical Marks: +1 / -0

The number of states in the minimum sized *DFA* that accepts the language defined by the regular expression

$$(0 + 1)^* (0 + 1)(0 + 1)^*$$

is _____.

19. GATE CSE 2014 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.3

If $L_1 = \{a^n \mid n \geq 0\}$ and $L_2 = \{b^n \mid n \geq 0\}$, consider

I) $L_1 \cdot L_2$ is a regular language

II) $L_1 \cdot L_2 = \{a^n b^n \mid n \geq 0\}$

Which one of the following is CORRECT?

- (A) Only (I)
- (B) Only (II)
- (C) Both (I) and (II)
- (D) Neither (I) nor (II)

20. GATE CSE 2014 Set 3

Numerical Marks: +1 / -0

The length of the shortest string NOT in the language (over $\Sigma = \{a, b\}$) of the following regular expression is

$$a^* b^* (ba)^* a^*$$

21. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the languages $L_1 = \phi$ and $L_2 = \{a\}$. Which one of the following represents $L_1 L_2^* U L_1^*$?

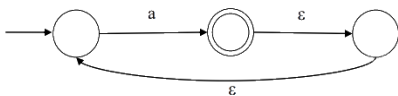
- (A) $\{\varepsilon\}$
- (B) ϕ
- (C) a^*
- (D) $\{\varepsilon, a\}$

22. GATE CSE 2012

MCQ (Single Correct Answer) Marks: +1 / -0.3

What is the complement of the language accepted by the *NFA* shown below?

Assume $\Sigma = \{a\}$ and ε is the empty string.



- (A) ϕ
- (B) $\{\varepsilon\}$
- (C) a^*
- (D) $\{a, \varepsilon\}$

23. GATE CSE 2010

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let L_1 recursive language. Let L_2 and L_3 be languages that are recursively enumerable but not recursive. Which of the following statement is not necessarily true?

- (A) $L_2 - L_1$ is recursively enumerable.
- (B) $L_1 - L_3$ recursively enumerable.
- (C) $L_2 \cap L_1$ is recursively enumerable.
- (D) $L_2 \cup L_1$ is recursively enumerable.

24. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which one of the following languages over the alphabet $\{0, 1\}$ is described by the regular expression $(0 + 1)^* 0(0 + 1)^* 0(0 + 1)^*$

- (A) The set of all strings containing the substring 00
- (B) The set of all strings containing at most two 0's
- (C) The set of all strings containing at least two 0's
- (D) The set of all strings that begin and end with either 0 or 1

25. GATE CSE 2003

MCQ (Single Correct Answer) Marks: +1 / -0.3

The regular expression $0^* (10^*)^*$ denotes the same set as

- (A) $(1^* 0)^* 1^*$
- (B) $0 + (0 + 10)^*$
- (C) $(0 + 1)^* 10 (0 + 1)^*$
- (D) None of the above.

26. GATE CSE 2003

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the set Σ^* of all strings over the alphabet $\Sigma = \{0, 1\}$. Σ^* with the concatenation operator for strings

- (A) Does not form a group
- (B) Forms a non-commutative group
- (C) Does not have a right identity element
- (D) Forms a group if the empty string is removed from Σ^*

27. GATE CSE 2001

MCQ (Single Correct Answer) Marks: +1 / -0.3

Given an arbitrary non-deterministic finite automaton (*NFA*) with N states, the maximum number of states in an equivalent minimized *DFA* is at least

- (A) N^2
- (B) 2^N
- (C) $2N$
- (D) $N!$

28. GATE CSE 2001

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the following two statements;

 $S_1 : \{0^{2n} \mid n \geq 1\}$ is a regular language $S_2 : \{0^m 1^n 0^{m+n} \mid m \geq 1 \text{ and } n \geq 1\}$ is a regular language

Which of the following statements is correct?

- (A) Only S_1 is correct
 (B) Only S_2 is correct
 (C) Both S_1 and S_2 are correct
 (D) None of S_1 and S_2 is correct

29. GATE CSE 2000

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let S and T be languages over $\Sigma = \{a, b\}$ represented by the regular expressions $(a + b)^*$ and $(a + b)^+$, respectively. Which of the following is true?

- (A) $S \subset T$
 (B) $T \subset S$
 (C) $S = T$
 (D) $S \cap T = \phi$

30. GATE CSE 2000

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let L denote the language generated by the grammar $S \rightarrow OS0 \mid 00$. Which one of the following is true?

- (A) $L = 0^+$
 (B) L is regular but not 0^+
 (C) L is context free but not regular
 (D) L is not context free

31. GATE CSE 1999

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the regular expression $(0 + 1)(0 + 1) \dots n$ times. The minimum state finite automation that recognizes the language represented by this regular expression contains:

- (A) n states
 (B) $n + 1$ states
 (C) $n + 2$ states
 (D) None of the above

32. GATE CSE 1998

MCQ (Single Correct Answer) Marks: +1 / -0.3

How many substrings of different lengths (non-zero) can be formed from a character string of length n ?

- (A) n
 (B) n^2
 (C) 2^n
 (D) $n(n + 1)/2$

33. GATE CSE 1998

MCQ (More than One Correct Answer) Marks: +1 / -0

The string 1101 does not belong to the set represented by

- (A) $110^*(0 + 1)$
 (B) $1(0 + 1)^*101$
 (C) $(10)^*(01)^*(00 + 11)^*$
 (D) $(00 + (11)^*0)^*$

34. GATE CSE 1998

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which of the following sets can be recognized by a Deterministic Finite-state Automation?

- (A) The numbers 1, 2, 4, 8, 2^n , written in binary.
 (B) The numbers 1, 2, 4, 2^n , written in unary.
 (C) The set of binary strings in which the numbers of zeros is the same as the numbers of ones.
 (D) The set $\{1, 101, 11011, 1110111, \dots\}$.

35. GATE CSE 1998

MCQ (Single Correct Answer) Marks: +1 / -0.3

If the regular set A is represented by $A = (01 + 1)^*$ and the regular set ' B ' is represented by $B = ((01)^*1^*)$, which of the following is true?

- (A) $A \subset B$
 (B) $B \subset A$
 (C) A and B are incomparable
 (D) $A = B$

36. GATE CSE 1997

MCQ (Single Correct Answer) Marks: +1 / -0.3

$\Sigma = \{a, b\}$, which one of the following sets is not countable.

- (A) Set of all strings over Σ
 (B) Set of all languages over Σ
 (C) Set of all regular languages over Σ
 (D) Set of all languages over Σ accepted by Turing Machines.

37. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let $L \subseteq \Sigma^*$ where $\Sigma = \{a, b\}$ which of the following is true?

- (A) $L = \{x \mid x \text{ has an equal number of } a's \text{ and } b's\}$ is regular
 (B) $L = \{a^n b^n \mid n \geq 1\}$ is regular
 (C) $L = \{x \mid x \text{ has more } a's \text{ than } b's\}$ is regular
 (D) $L = \{a^m b^n \mid m \geq 1, n \geq 1\}$ is regular

38. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which two of the following four regular expressions are equivalent?

(i) $(00)^*(\varepsilon + 0)$

(ii) $(00)^*$

(iii) 0^*

(iv) $0(00)^*$

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (i) and (iii)

(D) (iii) and (vi)

39. GATE CSE 1994

True or False Marks: +1 / -0

State True or False with one line explanation:

A FSM (Finite State Machine) can be designed to add two integers of any arbitrary length (arbitrary number of digits).

(A) TRUE

(B) FALSE
